

When I was in high school, I had a very talented English teacher who was influential in teaching me about writing. He really cared about the kids. I wouldn't say it was totally uncommon, but it certainly wasn't the norm. He had gone to Las Vegas on a trip. Afterward, I was over his house for dinner, and he mentioned that it was impossible to beat those guys.

That started me thinking about roulette. The ball seemed like a planet. I thought it might be possible to beat roulette by measuring the position and velocity of the ball and the rotor.

How old were you then?

I was 15.

So even at that age you apparently knew Newtonian physics.

When I was 16 they had a Southern California physics contest in which the best physics students competed. All the other students were 17 or 18 years old, but I placed first by a very wide margin.

You were self-taught in physics?

I taught myself in my own way. I was always an out-of-the-box thinker. I got hold of a college physics textbook and got through about two-thirds of it by the time of the physics contest, but that was apparently enough. I graduated when I was 16 and was able to attend college through a combination of scholarships and savings from my newspaper route. I graduated with a degree in physics from UCLA, and then I received a master's in physics. I had almost completed all the requirements for a PhD—I had done all the coursework, written exams, and orals—and was working on the last part of my thesis and ran into some math issues. It was a thesis on quantum mechanics.

I realized I needed to take more math courses. UCLA physics at the time was deficient in their math requirements. I basically had only the first two years of college math plus a few advanced math courses, but nothing close to what a math major would take. I had to jump into the graduate courses in math to get up to speed on the math I needed to finish the calculations in my thesis. Once I did that, I realized that considering the